

IMO Health's Response to the RFI on the Development of an Artificial Intelligence (AI) Action Plan

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Statement:

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Executive Summary

IMO Health is a leading provider of clinical terminology and healthcare data solutions, dedicated to enhancing the quality and interoperability of health information. We have considerable expertise in the development, use, and application of various types of AI to the clinical and lifescience research space. Our experience in these areas includes work using machine learning, deep learning, and LLM models and methods. In response to the request for information (RFI) on the Development of an Artificial Intelligence (AI) Action Plan, IMO Health emphasizes the critical importance of data enablement, robust terminology management, semantic alignment, and data provenance in healthcare AI applications. These elements are foundational to developing trustworthy, effective, and equitable AI systems that seamlessly integrate into clinical workflows and improve patient outcomes. IMO Health outlines below why we believe each element is essential for an effective AI Action plan.

1. Data Enablement in Healthcare AI

Healthcare data is inherently complex and multifaceted, encompassing a wide range of formats, sources, and structures. To harness this data for AI applications, it is imperative to implement comprehensive data enablement strategies that facilitate access, integration, and usability of health information. Key considerations include:

- **Data Integration:** Establishing interoperable systems that can aggregate data from diverse sources, including electronic health records (EHRs), laboratory systems, and patient-generated data, to create a unified and comprehensive dataset for AI model training.

- **Data Quality Assurance:** Implementing rigorous data governance frameworks to ensure the accuracy, completeness, and consistency of data, thereby enhancing the reliability of AI outputs. These frameworks relate to the data presented to AI models during their training phase to ensure accurate results that will not cause harm to patients. Similarly, data (including clinical and research recommendations) must be capable of being explained, and related back to clearly established and defined guidelines in peer-reviewed scientific and clinical evidence, and evidence evaluation frameworks.
- **Ethical Data Use:** Adhering to ethical standards and regulatory requirements to protect patient privacy and confidentiality, ensuring data use remains aligned with patient consent, privacy standards, and legal requirements.

2. Terminology Management and Semantic Alignment

Effective AI systems in healthcare rely on precise and consistent understanding of clinical concepts. Terminology management is pivotal in standardizing medical vocabulary, enabling semantic interoperability, and facilitating accurate data interpretation. Terminology management and semantic alignment have the following benefits for AI models:

- **Standardization of Clinical Terms:** Developing and maintaining comprehensive clinical terminologies that align with international standards, ensuring uniform interpretation of medical concepts across systems and stakeholders. This allows clear communications between participants in clinical care and clinical research, reducing errors, and allows for faster resolution of results.
- **Semantic Interoperability:** Implementing mapping strategies that translate disparate coding systems into a unified language, allowing seamless data exchange and integration across platforms. Semantic interoperability is essential for allowing AI systems to perform various types of logical reasoning, such as clinical inference, and to prevent the introduction of hallucinations and other forms of error into results.
- **Dynamic Updates:** Standard Terminologies require continuous updates to reflect the latest domain-specific context, regulatory standards, and best practices. Medical Terminologies encapsulate not only Standard Terminologies but also medical knowledge (e.g., diseases, procedures, labs, medications), clinical best practices, and innovations. Medical Terminologies ensure that AI systems operate with current and relevant information and utilize a robust source of truth with data provenance to train upon.

3. Preparing Healthcare Data for AI Model Training

The high dimensionality and variability of healthcare data necessitate meticulous preprocessing to render it suitable for AI model development. Essential steps include:

- **Data Cleaning:** Identifying and rectifying errors, inconsistencies, and redundancies in datasets to enhance data integrity and model performance.
- **Feature Engineering:** Selecting and transforming relevant data attributes to improve model accuracy and predictive capabilities.

- **Handling Missing Data:** To maintain dataset robustness, appropriate techniques, such as imputation methods or leveraging auxiliary information, should be employed to address gaps in data.

4. Provenance and Transparency in AI Models

Ensuring the traceability and transparency of AI models is crucial for building trust among clinicians and patients. IMO Health advocates for:

- **Clear Documentation:** Maintaining detailed records of data sources, preprocessing steps, and model development processes to facilitate reproducibility and accountability.
- **Explainability:** Designing AI models that provide interpretable outputs, enabling clinicians to understand and trust AI-driven recommendations and decisions.
- **Continuous Monitoring:** Implementing systems to track AI model performance over time, ensuring sustained accuracy, early detection of model drift, and proactive bias mitigation.

5. Additional Considerations

To further enhance the development and deployment of AI in healthcare, IMO Health recommends an AI action plan include the following:

- **AI Governance Frameworks:** Establishing policies and oversight mechanisms that guide the ethical development and application of AI, addressing issues such as bias, fairness, and accountability.
- **Collaboration and Stakeholder Engagement:** Fostering partnerships among healthcare providers, technology developers, policymakers, and patients to co-create AI solutions aligned with clinical needs and societal values.
- **Investment in Education and Training:** Developing programs to equip healthcare professionals with the knowledge and skills to effectively utilize AI tools, ensuring that technological advancements translate into improved patient care.

Conclusion

IMO Health is committed to advancing AI in healthcare through robust data enablement, precise terminology management, and unwavering dedication to data quality and transparency. By addressing these critical areas, the forthcoming AI Action Plan can lay a strong foundation for AI innovations that are ethical, effective, and centered on enhancing patient outcomes.